
Credit Valuation Adjustment for Derivative Contracts

Counterparty Credit Risk Modelling, Monte-Carlo Exposure
Simulation and CDS-Implied Default Probabilities

Project scope: Credit Valuation Adjustment for derivative contracts

Counterparty: Deutsche Bank AG (unilateral CVA)

Calculation date: 31 December 2025

Tools used: R, Microsoft Excel

Course: Quantitative Finance

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Executive summary

- We measured the counterparty credit risk on a small portfolio of two trades held against Deutsche Bank: a 3-year EUR/PLN FX forward and a 3-year EUR receiver interest rate swap.
 - Future exposures were built with Monte-Carlo simulation, using 10,000 scenarios and monthly time steps over the 3-year life.
 - The counterparty's default probabilities were taken from market CDS spreads.
 - On their own, the two trades carry a CVA of **3,126 PLN** (FX forward) and **9,699 PLN** (swap).
 - Netting the trades together brings the portfolio CVA down to **10,485 PLN**, a saving of about **18%**.
 - Posting monthly variation margin cuts the charge much further, to **1,942 PLN**, a reduction of about **82%**.
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1 Introduction

Whenever two parties sign an over-the-counter (OTC) derivative, each side takes on *counterparty credit risk*. This is simply the risk that the other party goes bankrupt before the contract ends and cannot pay what it owes. A derivative is not like a normal loan: its value goes up and down every day as the market moves, and it can be worth money to either side. So the amount we could lose on a default is not fixed, and we have to model it. The standard way to put a price on this risk is the **Credit Valuation Adjustment (CVA)**, which is the expected loss from the counterparty defaulting. We subtract it from the risk-free value of the trade:

$$\text{Risk-adjusted price} = \text{Risk-free price} - \text{CVA}.$$

In this project we compute the *unilateral* CVA (we only worry about the counterparty defaulting, not ourselves) for a portfolio held against **Deutsche Bank** as of 31 December 2025. The portfolio has two instruments, set by the project brief:

- a **Buy-EUR / Sell-PLN FX forward**: 3-year maturity, notional 1,000,000 EUR, strike $K = 4.4439$;
- a **EUR receiver interest-rate swap** on EURIBOR 3M: 3-year maturity, notional 10,000,000 EUR, fixed rate 2.02% received annually.

The rest of the report follows the natural order of the calculation: the theory (Section 2), the two instruments and how we value them (Section 3), the data and calibration (Section 4), the exposure simulation (Section 5), the CDS bootstrap that gives the default probabilities (Section 6), the CVA results (Section 7) and the variation-margin study (Section 8), then the conclusions.

Portfolio CVA 10,485 PLN	Netting benefit 18.2%	CVA after VM 1,942 PLN
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2 Theoretical Framework

2.1 From counterparty risk to CVA

A derivative is worth a positive amount to us when the counterparty owes us money (it is an *asset*) and a negative amount when we owe the counterparty (a *liability*). We can only lose money on a default when the trade is an asset to us, so the credit exposure is the positive part of the mark-to-market (MtM) value, $\max(\text{MtM}, 0)$. If we average this over many simulated futures, we get the **Expected Exposure (EE)** at each future date. In continuous time the unilateral CVA brings together the loss given default, the discounted expected exposure and the chance of default:

$$\text{CVA}(t) = \text{LGD} \int_t^T \text{EE}(u) d\text{PD}_C(u), \quad (1)$$

where $\text{LGD} = 1 - R$ is the loss given default, $\text{EE}(u)$ is the present value of expected exposure, and $\text{PD}_C(u)$ is the counterparty's default probability. We use the same simplifying assumptions as the brief: no wrong-way risk, exposure and default are independent, and we treat each contract on its own (unilateral).

2.2 The discrete CVA formula

To actually compute it, we turn the integral into a sum over the monthly grid $t_0 < t_1 < \dots < t_N$ (here $N = 36$ months and $\Delta t = 1/12$):

$$\text{CVA} \approx (1 - R) \sum_{i=1}^N \text{EE}(t_i) \cdot \text{DF}(0, t_i) \cdot q(t_{i-1}, t_i) \quad (2)$$

where $DF(0, t_i)$ is the risk-free discount factor to time t_i , $q(t_{i-1}, t_i) = PS(t_{i-1}) - PS(t_i)$ is the chance the counterparty defaults inside the month $(t_{i-1}, t_i]$, PS is the survival probability, and R is the recovery rate. The nice thing is that the three pieces are built separately: the EE comes from simulating the market (Section 5), the q values come from the CDS bootstrap (Section 6) and the discount factors come from the rate curve. Equation (2) is the engine behind the whole project.

3 Instruments: Mechanics and Valuation

3.1 EUR/PLN FX forward

The forward fixes the price at which we will buy 1,000,000 EUR with PLN in three years, at the strike $K = 4.4439$. Its payoff at maturity is just the notional times the gap between the final spot and the strike, $N_{fx}(S_T - K)$. At any earlier date t we value it with the Garman-Kohlhagen forward rule: we discount the EUR side at the EUR rate and the PLN side at the PLN rate over the time left $\tau = T - t$:

$$V_{FX}(t) = N_{fx} \left[S(t) e^{-r_{EUR}\tau} - K e^{-r_{PLN}\tau} \right]. \quad (3)$$

Since we are long EUR, the trade gains value when the zloty weakens (EUR/PLN goes up) and loses value when it strengthens. The profile is roughly a straight line in the spot.

3.2 EUR receiver interest-rate swap

The swap pays us a fixed annual coupon (2.02% on 10,000,000 EUR) and we pay quarterly EURIBOR 3M in return. To mark it to market we value both legs:

$$V_{IRS}(t) = \sum_{T_k > t} [\text{Fixed}(T_k) - \text{Floating}(T_k)] \cdot \frac{P_{EUR}(0, T_k)}{P_{EUR}(0, t)}. \quad (4)$$

Simplifying the floating leg. One thing the brief asks us to show is that the floating leg can be written in a much simpler way. A floating coupon that is set at the start of the period and paid at the end, $N \cdot L(T_{k-1}, T_k) \Delta t$, has a present value equal to the difference of two zero-coupon bonds, $N[P(0, T_{k-1}) - P(0, T_k)]$. When we add up the whole schedule the middle terms cancel out (they telescope), and the entire floating leg simplifies to

$$PV_{\text{float}} = N[P(0, T_{\text{start}}) - P(0, T_{\text{end}})]. \quad (5)$$

So the floating leg does not actually need a forecast of future rates, only two discount factors. For the exposure simulation, though, we keep the full cash-flow version so that the spread of future swap values still reflects the simulated EURIBOR paths.

4 Market Data and Model Calibration

Where the data comes from. We use four pieces of market data:

- **EUR/PLN spot history** from Stooq, covering 2023 to 2025 (766 daily points);
- **EURIBOR 3M / ESTR history** from the ECB Data Portal over the same period;
- **discount factors** built from flat EUR and PLN money-market rates;
- **CDS quotes** for Deutsche Bank, which give us the credit curve.

We use the historical series only to *calibrate* the volatilities and to check the FX / interest-rate correlation. All of the actual pricing is then done under the risk-neutral drift. **How each factor moves.** We model the FX rate with a *Geometric Brownian Motion* (GBM). GBM keeps the rate positive, which is what we want for an exchange rate, and it has a clean one-step formula:

$$S_{t+1} = S_t \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z \right], \quad \mu = r_{PLN} - r_{EUR}. \quad (6)$$

We model the short rate behind the swap with an *Arithmetic Brownian Motion* (ABM), $r_{t+1} = r_t + \mu \Delta t + \sigma \sqrt{\Delta t} Z$. ABM is a good choice here because it allows negative rates, which are realistic for the EUR market. **We simulated the two factors independently.** The brief suggested joining the two processes with a Cholesky decomposition so that they share their correlation. We first measured the historical correlation between FX returns and EURIBOR changes, and it came out at only **0.014**, which is basically zero. Because the two risk factors hardly move together, coupling them would not change the answer in any meaningful way. We therefore kept the model simple and simulated the FX rate (GBM) and the interest rate (ABM) on their own. This gives the same result as a joint simulation, but it is easier to build and to explain.

Table 1. Calibrated model parameters (calculation date 31.12.2025).

Parameter	Value	Parameter	Value
Horizon T / steps	3 yr / 36 monthly	EUR rate r_{EUR}	2.00%
Simulations	10,000 paths	PLN rate r_{PLN}	3.75%
FX notional N_{fx}	1,000,000 EUR	FX strike K	4.4439
IRS notional N_{irs}	10,000,000 EUR	Swap fixed rate	2.02%
Recovery R / LGD	40% / 60%	FX/IR correlation	0.014 (negligible)

5 Exposure Simulation

For each of the 10,000 scenarios we move the risk factors forward month by month, revalue every trade along the path, keep the positive part of the value, and then average across all paths. That average is the EE profile, $EE(t_i) = \frac{1}{M} \sum_m \max(\text{MtM}_m(t_i), 0)$.

5.1 FX forward exposure

We start each path from today's spot rate and step it forward with the GBM formula, then revalue the forward at every step with the Garman-Kohlhagen rule. Because the simulated rate has more and more room to move as time passes, the exposure first grows, and then it shrinks back toward maturity as the remaining life of the trade runs out.

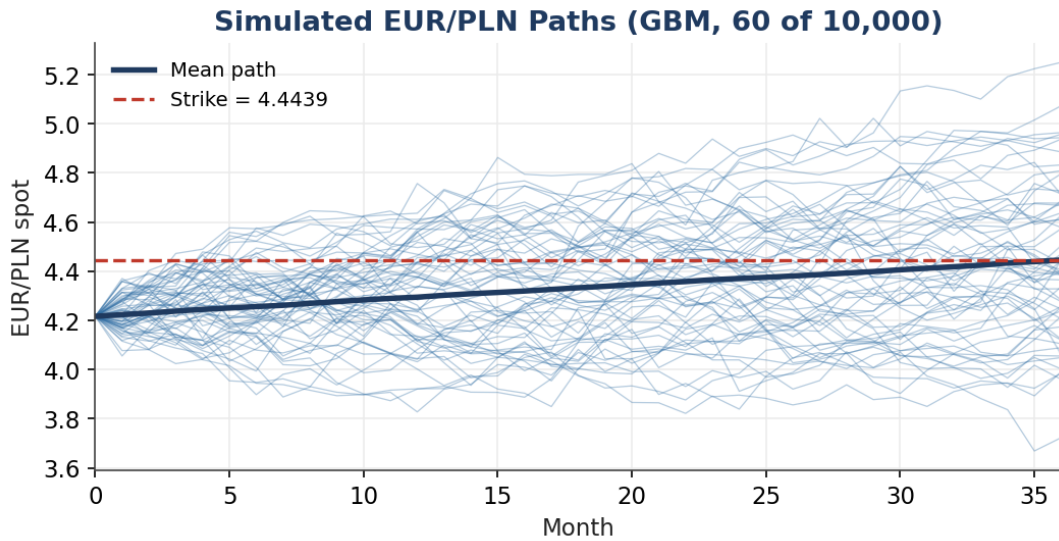


Figure 1. Simulated EUR/PLN spot paths under GBM (60 of 10,000 shown).

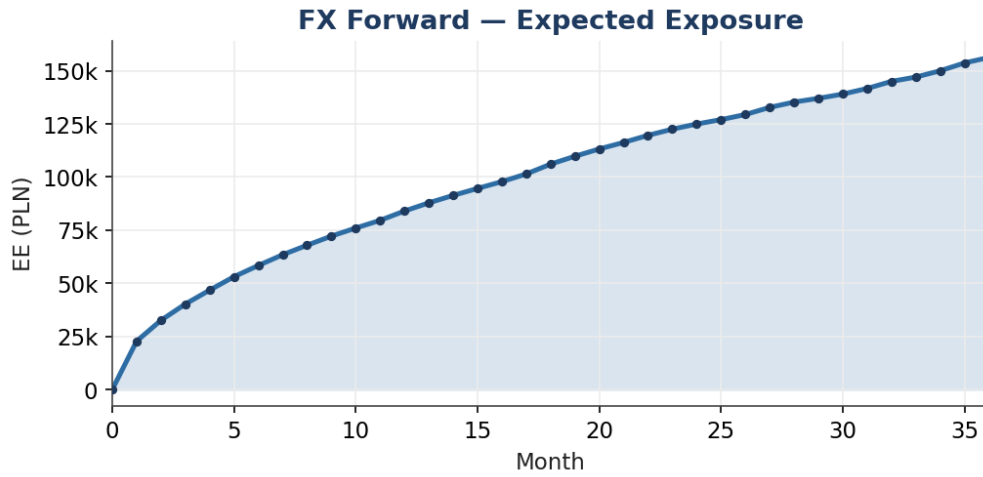


Figure 2. FX forward expected-exposure profile.

5.2 Interest-rate swap exposure

We simulate EURIBOR with the ABM formula and value the swap by discounting its remaining fixed and floating cash flows at each step. Because we receive the fixed coupons and pay floating, the swap tends to drift into the money as the contract runs, so its exposure keeps building over the life rather than dying away. The swap also carries ten times the notional of the FX forward, so it is the larger of the two exposures.

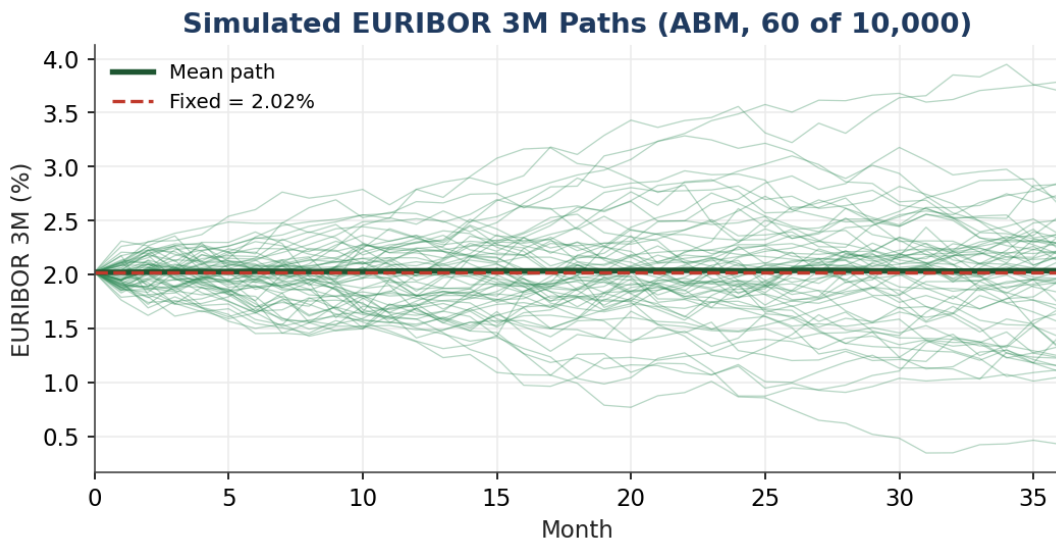


Figure 3. Simulated EURIBOR 3M paths under ABM (60 of 10,000 shown), spreading around the 2.02% fixed rate. Low volatility keeps the paths close together.

5.3 Portfolio exposure and netting

With a netting agreement we add the two MtMs together *before* taking the positive part, so a gain on one trade can offset a loss on the other: $EE_{\text{port}}(t) = \mathbb{E}[\max(\text{MtM}_{FX} + \text{MtM}_{IRS}^{PLN}, 0)]$. Figure 5 puts the three profiles on one chart. The portfolio exposure is smaller than the two standalone profiles added together, and that difference is the netting benefit we measure in Section 7.

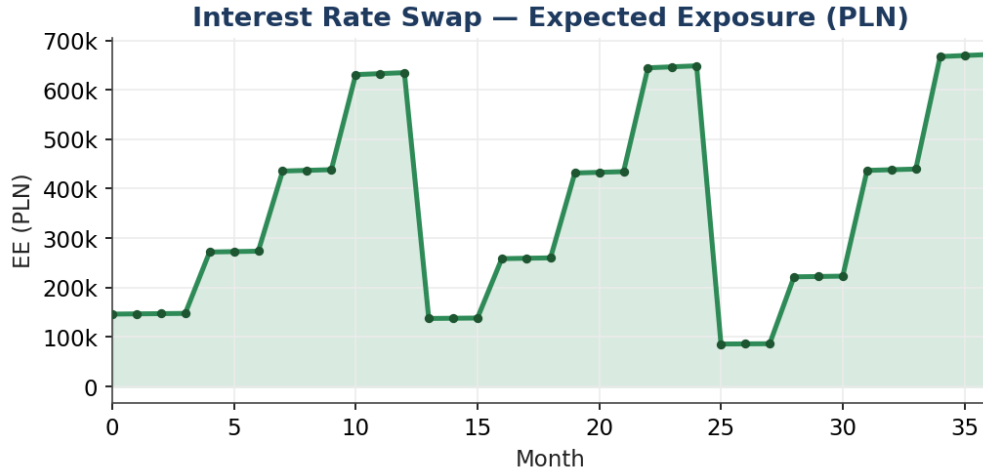


Figure 4. IRS expected-exposure profile in PLN. The steps line up with the annual fixed-coupon dates of the receiver swap.

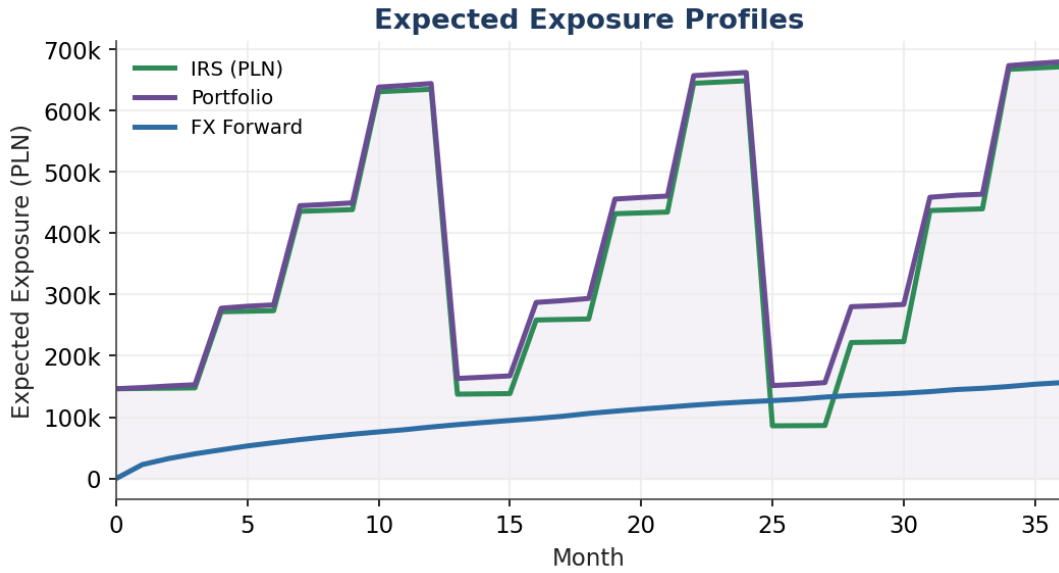


Figure 5. Expected-exposure profiles for the FX forward, the IRS and the netted portfolio (all in PLN). The portfolio profile is smaller than the two added together because of netting.

6 Implied Default Probabilities: CDS Bootstrap

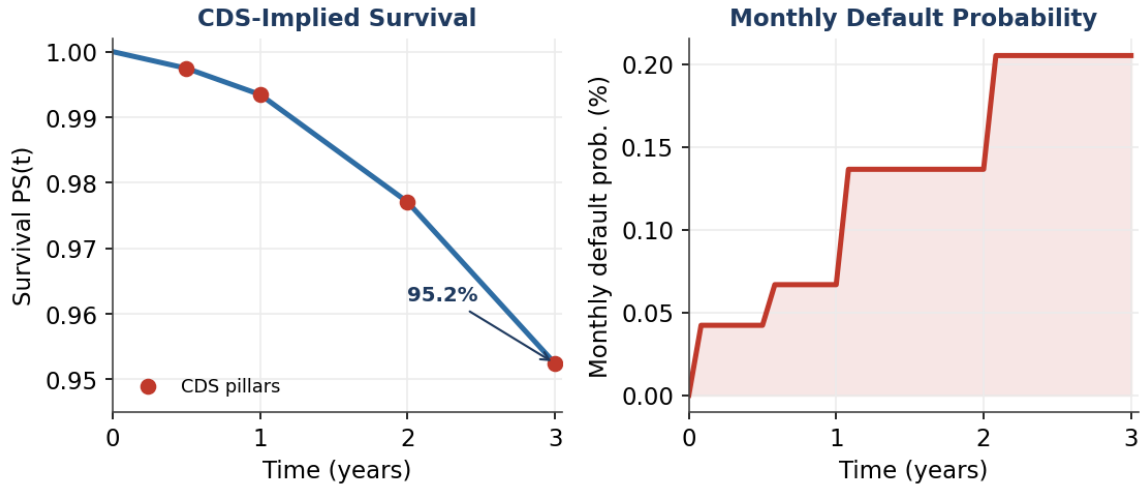
The default part of the CVA needs the counterparty's survival curve, and we get it from Deutsche Bank's quoted CDS spreads. A CDS works like insurance: the protection buyer pays a regular premium, and if the reference name defaults the protection seller pays out $(1 - R)$. If we set the two legs equal at the start of the contract, we get the par-spread formula:

$$S_N = \frac{\text{LGD} \sum_{n=1}^N \text{DF}(0, T_n) [\text{PS}(T_{n-1}) - \text{PS}(T_n)]}{\sum_{n=1}^N \text{DF}(0, T_n) \text{PS}(T_n) \Delta t_n}. \quad (7)$$

We solve this one pillar at a time. Once we know the survival probabilities up to tenor T_{N-1} , the quote S_N lets us solve for the next one, $\text{PS}(T_N)$. That step-by-step process is what **bootstrapping** means. We start from the closed-form first pillar, $\text{PS}(T_1) = \text{LGD}/(\text{LGD} + S_1 T_1)$, and find each later pillar with a simple root search. We then interpolate the curve onto the monthly grid and take differences to get the monthly default probabilities $q(t_{i-1}, t_i)$ used in Equation (2). The code is in Appendix A.

Table 2. Input CDS spreads and bootstrapped survival / cumulative-default probabilities.

Tenor	CDS spread	Survival PS	Cum. default 1–PS
0.5 yr	30.5 bp	0.99747	0.25%
1 yr	39.4 bp	0.99346	0.65%
2 yr	69.4 bp	0.97707	2.29%
3 yr	96.7 bp	0.95243	4.76%

**Figure 6.** Left: the CDS-implied survival curve with the four calibration pillars. Right: the monthly default probability $q(t)$ obtained by differencing the survival curve.

The bootstrapped curve implies a cumulative default probability of about 4.8% over the three-year horizon. Because the survival curve falls a little faster at longer tenors, the monthly default probability is slightly larger later on, so later exposures carry a little more weight in the CVA sum.

7 CVA Results

Putting the EE profiles, the discount factors and the monthly default probabilities into Equation (2) gives the CVA for each trade and for the netted portfolio (Table 3 and Figure 7).

Table 3. CVA results (LGD = 60%, calculation date 31.12.2025).

Instrument / level	CVA (PLN)	Comment
FX forward (standalone)	3,125.90	smaller notional, humped EE
IRS in PLN (standalone)	9,699.28	larger: 10x notional, rising EE
Portfolio (netted)	10,484.91	below the standalone sum (12,825)
Portfolio with monthly VM	1,942.39	only the residual gap risk is left

The netting benefit. On their own the two CVAs add up to 12,825 PLN, while the netted portfolio CVA is 10,485 PLN. The saving from netting is

$$1 - \frac{CVA_{\text{port}}}{CVA_{FX} + CVA_{IRS}} = 1 - \frac{10,485}{12,825} \approx 18.2\%.$$

The benefit is real but modest, which fits the inputs: the FX and interest-rate factors are almost uncorrelated, and the swap exposure is much larger than the FX exposure, so there is limited room for the two to offset. Figure 8 shows the same CVA built up step by step along the monthly grid, which is exactly how Equation (2) sums the per-month contributions.

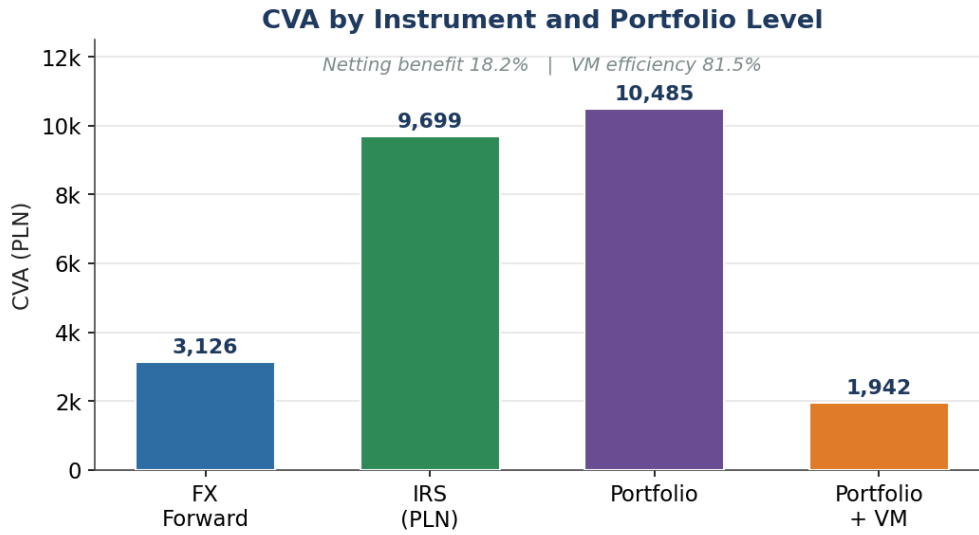


Figure 7. CVA by trade and at portfolio level. Netting lowers the charge compared with the two standalone numbers added together, and variation margin lowers it further still.

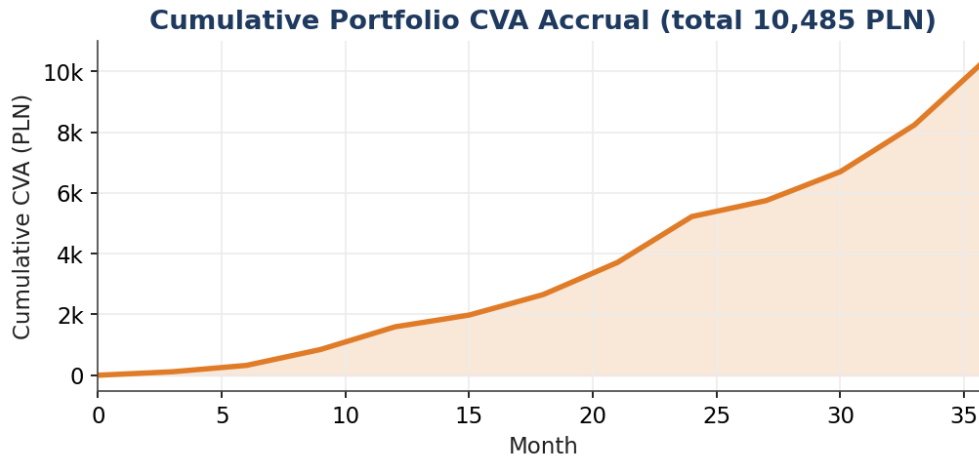


Figure 8. Cumulative build-up of the netted portfolio CVA over the 3-year horizon.

8 Impact of Variation Margin

Variation margin (VM) is collateral that the two sides exchange to keep up with the portfolio's value. With VM in place, if the counterparty defaults only the change since the last margin call is at risk, not the whole exposure. The brief sets VM in PLN, exchanged every month, with

$$VM(T) = MtM_{IRS}(T-1) \cdot FX(T-1) + MtM_{FX}(T-1), \quad VM(0) = 0. \quad (8)$$

The collateralised exposure is whatever is left after subtracting the posted margin, $\max(MtM_{port}(T) - VM(T), 0)$. Since the margin equals the previous month's value, it cancels the level of the portfolio and only the one-month change in value is still exposed. This residual is reset at every monthly call, which is why the margined exposure stays small throughout (Figure 9). The CVA drops from 10,485 PLN to 1,942 PLN, a **VM efficiency** of

$$1 - \frac{CVA_{VM}}{CVA_{port}} = 1 - \frac{1,942}{10,485} \approx 81.5\%.$$

The remaining 18.5% is the gap risk that builds up between monthly calls and cannot be removed by monthly margining. Calling margin more often (for example weekly or daily) would reduce this residual further.

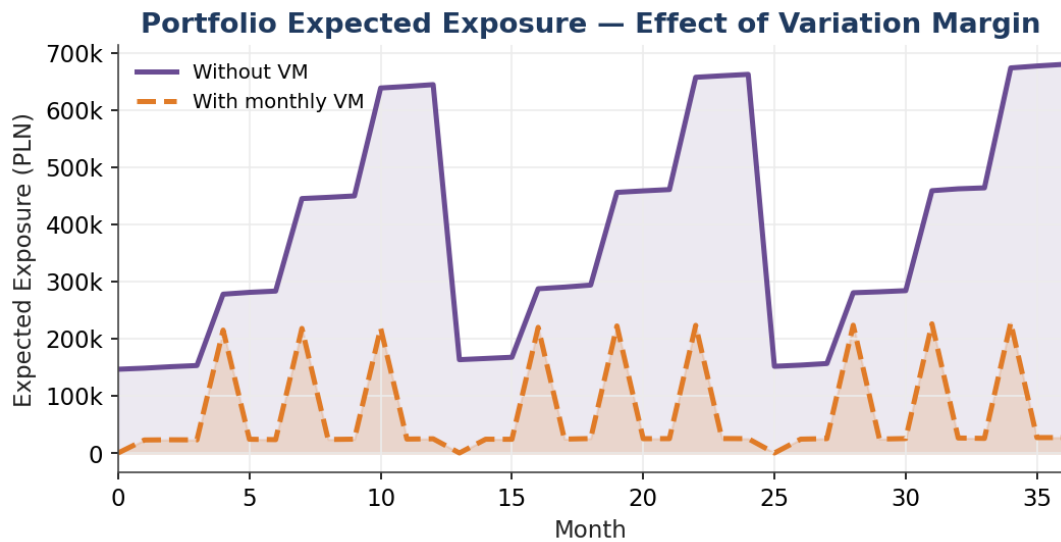


Figure 9. Portfolio expected exposure without VM (solid) and with monthly VM (dashed). Margining removes the level of the exposure and leaves only the monthly gap risk.

9 Conclusions

- **The swap drives the credit charge.** Its standalone CVA (9,699 PLN) is about three times the FX forward's (3,126 PLN), mainly because it has ten times the notional and an exposure profile that keeps rising.
- **Netting helps, but only a little.** The portfolio CVA (10,485 PLN) is 18.2% below the two standalone numbers added together. The benefit is limited by the near-zero FX / interest-rate correlation and the size gap between the two trades.
- **Correlation did not matter here.** The historical FX / interest-rate correlation is only 0.014, so we simulated the two processes independently (GBM for the FX rate, ABM for the interest rate) instead of coupling them with a Cholesky decomposition. A joint simulation would give essentially the same CVA.
- **Variation margin is the strongest tool.** Monthly PLN margining removes 81.5% of the portfolio CVA, leaving only 1,942 PLN of monthly gap risk, which more frequent margin calls would reduce further.

Overall, the project walks through the whole CVA pipeline from start to finish: calibrate to market data, simulate the exposures with Monte-Carlo, bootstrap the default probabilities from CDS spreads, and combine everything into a credit charge. We built it in two ways, once in R and once in an Excel workbook, and the two agree on the numbers reported above.

A Key Code: CDS Bootstrap and CVA

The two most important pieces of the model are the recursive CDS bootstrap and the CVA sum, shown below.

```
## CDS bootstrap: imply survival probabilities from quoted spreads
cdsTenors <- c(0.5, 1, 2, 3)
cdsSpreads <- c(0.00305, 0.00394, 0.006935, 0.00967)
DF <- exp(-r_eur * cdsTenors); RR <- 0.40; LGD <- 1 - RR
PS <- numeric(length(cdsTenors))
solve_PS <- function(k, PS_prev) { # root-solve pillar k
  S <- cdsSpreads[k]
  f <- function(PS_k) {
    PS_full <- c(1, PS_prev, PS_k); PL <- 0; DL <- 0
    for (i in 2:(k + 1)) {
```

```

    dt_cds <- cdsTenors[i-1] - ifelse(i==2, 0, cdsTenors[i-2])
    DL <- DL + LGD * DF[i-1] * (PS_full[i-1] - PS_full[i]) # default leg
    PL <- PL + S * DF[i-1] * PS_full[i] * dt_cds # premium leg
  }
  PL - DL # = 0 at the fair spread
}
uniroot(f, c(0.5, 1))$root
}
PS[1] <- LGD / (LGD + cdsSpreads[1] * cdsTenors[1])
for (k in 2:length(cdsTenors)) PS[k] <- solve_PS(k, PS[1:(k-1)])
default_prob <- c(0, head(PS_interp,-1) - tail(PS_interp,-1)) # monthly q(t)
## FX and rate are simulated independently (correlation ~ 0.014)
## FX: Geometric Brownian Motion / Rate: Arithmetic Brownian Motion
## CVA: combine EE, discount factors and default probabilities
CVA_FX <- LGD * sum(fx_EE * discount_factors * default_prob)
CVA_IRS_PLN <- LGD * sum(irs_EE_pln * discount_factors * default_prob)
CVA_PORTFOLIO <- LGD * sum(portfolio_EE * discount_factors * default_prob)
CVA_VM <- LGD * sum(vm_EE * discount_factors * default_prob)
netting_benefit <- 1 - CVA_PORTFOLIO / (CVA_FX + CVA_IRS_PLN) # 18.2%
vm_efficiency <- 1 - CVA_VM / CVA_PORTFOLIO # 81.5%

```

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